

FedTinyRT — Project Brief (Sleep Apnea)

REALITY FILTER. Tags: [Verified] = checked against a source (Renesas EK-RA8P1 product page, the FSP-generated board pin config in this repo, or a cited paper/dataset page) · [Inference] = reasoned from verified facts · [Unverified] = not confirmed · [Decision] = chosen by us, not a fact about the world. Nothing here is a clinical claim.

Scope note: this brief describes the **sleep-apnea direction**. SYSTEM_OVERVIEW.md, context.md, and RFC-001-systems-contract.md still describe the earlier **Parkinson's Freezing-of-Gait** target and have **not** been rewritten yet — where they conflict with this file, treat them as the old target. [Verified – read in this repo]

What we are building

FedTinyRT is a privacy-preserving, on-device screening system for **obstructive sleep apnea (OSA)** built on the **Renesas EK-RA8P1** (Cortex-M85 @ 1 GHz + Cortex-M33 @ 250 MHz + Ethos-U55 NPU) running **µT-Kernel 3.0**. One board sits at the bedside of a sleep clinic or ward as a base station. Through the night it ingests several cheap, low-burden signals at once — breathing and snoring **audio** from the board's onboard **PDM MEMS microphones**, **SpO₂/PPG** from an I²C pulse-oximeter module, and **body-position / respiratory-effort motion** from an accelerometer — windows them into fixed epochs, and runs an **INT8 multimodal neural network on the Ethos-U55 NPU** to flag apnea/hypopnea events in real time and accumulate an apnea-hypopnea index (AHI) estimate. Raw waveforms, features, and labels never leave the board. [Decision]

The second half of the system is **federated learning**. Each clinic's board fine-tunes the model's head **on its own patients, on-device, on the M33 core** while the M85 keeps screening in real time on physically separate silicon — then sends **only FP32 weights** over wired Ethernet to an aggregator that averages them (**FedAvg**) into a new global model and sends it back. No recording, no oximetry trace, and no clinician label ever crosses the network; the M33 is the egress gate that enforces this. [Decision] The intended contribution is the **embedded systems realization** — multimodal apnea screening + on-device training + cross-site federation on an RTOS-driven dual-core MCU with a measured real-time guarantee — not a new ML algorithm, and not a validated medical device. We do **not** claim federation "fixes," "guarantees," or "ensures" cross-site performance; it is an experiment, reported with uncertainty. [Inference – positioning, based on our own literature read]

Which part of the board does what [Decision]

Hardware facts are [Verified] (Renesas EK-RA8P1 kit page + this repo's bsp_pin_cfg.h); the use we put each part to is our plan, not a measured result.

Part	What it is	What we use it for
Cortex-M85 @ 1 GHz + Helium	fast core, SIMD DSP	Hard-real-time sensor ingest (audio, SpO ₂ , accel), DMA ring buffers, CMSIS-DSP front end (log-mel / FFT), epoch alignment across modalities, issues NPU inference, drives the display.
Ethos-U55 NPU @ 500 MHz, 256 GOPS	INT8/INT16 NN accelerator, inference only	Forward pass of the multimodal apnea model, per epoch. Never backprop . Ops Vela can't take fall back to M85 CMSIS-NN.
Cortex-M33 @ 250 MHz	slower second core, no NPU/Helium	Background on-device training of the classifier head (FP32), federation client (store-and-forward), and the egress gate : weights may leave, raw data may not.
M85 TCM (256 KB)	tightest-latency RAM	Audio/PPG/accel DMA ring buffers + hottest DSP working set.
M33 TCM (128 KB)	second-core scratch	Trainable head weights, gradients, Adam state, one micro-batch.

Part	What it is	What we use it for
		Budget ≤ 96 KB — not yet benchmarked . [Unresolved]
2 MB SRAM (ECC)	on-chip data	NN activations the NPU streams + per-modality feature buffers.
1 MB MRAM	on-chip non-volatile	Firmware + A/B model slots — new global model written to the inactive slot, checksummed, then atomically switched.
64 MB SDRAM	external bulk RAM	Multi-minute epoch buffers and overnight event/feature history for the AHI tally.
64 MB Octo-SPI Flash	external bulk non-volatile	Dataset replay for the demo (PSG recordings fed into the live pipeline), event logs, calibration.
Onboard PDM MEMS mic	MIC_CLK / MIC_DAT	Modality 1: snoring / breathing sound, gasp and arousal onsets.
I²C (Pmod / Grove / Qwiic / I3C)	sensor bus	Modality 2: SpO₂ + PPG pulse-oximeter module — desaturation and pulse-rate surge. Part not chosen. [Unverified]
I²C or SPI Pmod	sensor bus	Modality 3: accelerometer — body position + chest/abdomen respiratory effort (obstructive vs central).
Arduino analog pins (ADC)	ARDUINO_AN0...AN5	Reserved for an optional ECG / nasal-airflow analog front end. Not designed. [Speculation]
Ethernet RGMII (RJ45)	only network link on the board	Federation only : FP32 weights out, Global_Model[r] in, during sync windows. PHY off otherwise.
MIPI display (7" 1024×600 touch)	kit expansion board	Clinician view: live respiration/SpO ₂ trace, flagged events, running AHI, session start/stop.
Camera connector (OV5640)	kit expansion board	Contactless chest-motion idea. Out of v1 scope — privacy cost, and it fights the parallel-LCD pins on SW4. [Speculation]
USB High-Speed	USB-C	Bulk export of anonymized session summaries and benchmark dumps to the dev PC.
Onboard J-Link + VCOM UART	debug	Telemetry, deadline-miss counters, on-target debug (SEGGER RTT).
User LEDs / buttons	3 LEDs, 2 buttons	LED = recording / liveness / fault state; buttons = start and stop a night session.
TrustZone	M85 security extension	Deferred: optional hardware-backed egress gate, only if v1 lands early.
Why the split matters: inference (M85 + NPU) and training (M33) run on physically separate cores at the same time , so on-device learning can never delay real-time screening. That is the RTOS thesis of the project. [Decision — the 0-missed-deadline claim must still be measured]		